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✓ A. For olympics dataset:(tab separated file)

```
!pip install -q pyspark
```

```
from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \
    .master("local[*]") \
    .appName("ColabSpark") \
    .getOrCreate()

print("Spark Version:", spark.version)

!pyspark --version
```

Spark Version: 4.0.2
 WARNING: Using incubator modules: jdk.incubator.vector
 Welcome to



Using Scala version 2.13.16, OpenJDK 64-Bit Server VM, 17.0.18
 Branch HEAD
 Compiled by user runner on 2026-02-02T08:08:13Z
 Revision 7cc3b9bcd8a8c923f23cd9ce922530e1becf1
 Url <https://github.com/apache/spark>
 Type --help for more information.

1. Print schema of the dataset

```
df= spark.read.csv("olympix_data.csv",sep=";", header=True, inferSchema=True)
```

```
print("=== 1. SCHEMA ===")
df.printSchema()
```

```
=== 1. SCHEMA ===
root
 |-- Athlete: string (nullable = true)
 |-- Age: integer (nullable = true)
 |-- Country: string (nullable = true)
 |-- Year: integer (nullable = true)
 |-- Closing Ceremony Date: string (nullable = true)
 |-- Sport: string (nullable = true)
 |-- Gold Medals: integer (nullable = true)
 |-- Silver Medals: integer (nullable = true)
 |-- Bronze Medals: integer (nullable = true)
 |-- Total Medals: integer (nullable = true)
```

2. Show only first 5 rows

```
df.show(5)
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Athlete|Age|Country|Year|Closing Ceremony Date|Sport|Gold Medals|Silver Medals|Bronze Medals|Total Medals|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Michael Phelps|23|United States|2008|08-24-08|Swimming|8|0|0|8|
| Michael Phelps|19|United States|2004|08-29-04|Swimming|6|0|2|8|
| Michael Phelps|27|United States|2012|08-12-12|Swimming|4|2|0|6|
| Natalie Coughlin|25|United States|2008|08-24-08|Swimming|1|2|3|6|
| Aleksey Nemov|24|Russia|2000|10-01-00|Gymnastics|2|1|3|6|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
```

3. Select specific columns (Athlete, Country, Sport)

```
df = df.select("Athlete", "Country", "Sport")
df.show(5)
```

```
+-----+-----+-----+
| Athlete|Country|Sport|
+-----+-----+-----+
```

```
+-----+-----+-----+
| Michael Phelps|United States| Swimming|
| Michael Phelps|United States| Swimming|
| Michael Phelps|United States| Swimming|
|Natalie Coughlin|United States| Swimming|
| Aleksey Nemov| Russia|Gymnastics|
+-----+-----+-----+
only showing top 5 rows
```

4. Filter athletes with more than 5 total medals(Write sql query)

```
df_full.createOrReplaceTempView("olympics")
result = spark.sql(sql_query)
result.show()
```

```
+-----+-----+-----+
| Athlete| Country| Sport|
+-----+-----+-----+
| Michael Phelps|United States| Swimming|
| Michael Phelps|United States| Swimming|
| Michael Phelps|United States| Swimming|
|Natalie Coughlin|United States| Swimming|
| Aleksey Nemov| Russia|Gymnastics|
+-----+-----+-----+
```

5. Count medals won by each country

```
from pyspark.sql.functions import sum, col

print("\n=== 5. MEDALS BY COUNTRY (Top 5) ===")
medals_by_country = df_full.groupBy("Country") \
    .agg(sum("`Total Medals`").alias("Sum_Total_Medals")) \
    .orderBy(col("Sum_Total_Medals").desc())
medals_by_country.show(5)
```

```
=== 5. MEDALS BY COUNTRY (Top 5) ===
+-----+-----+
| Country|Sum_Total_Medals|
+-----+-----+
|United States| 1312|
| Russia| 768|
| Germany| 629|
| Australia| 609|
| China| 530|
+-----+-----+
only showing top 5 rows
```

6. Average age of athletes per sport

```
from pyspark.sql.functions import avg

print("\n=== 6. AVG AGE PER SPORT ===")
avg_age = df_full.groupBy("Sport") \
    .agg(avg("Age").alias("Average_Age"))
avg_age.show(5)
```

```
=== 6. AVG AGE PER SPORT ===
+-----+-----+
| Sport| Average_Age|
+-----+-----+
| Gymnastics| 21.34020618556701|
| Tennis| 26.676056338028168|
| Boxing| 24.20212765957447|
|Short-Track Speed...| 22.697916666666668|
| Ice Hockey| 26.9296875|
+-----+-----+
only showing top 5 rows
```

7. Maximum gold medals won by a single athlete

```
from pyspark.sql.functions import col, sum

print("\n=== 7. MAX GOLD MEDALS (Single Athlete) ===")
athlete_golds = df_full.groupBy("Athlete") \
    .agg(sum("`Gold Medals`").alias("Total_Gold_Medals")) \
    .orderBy(col("Total_Gold_Medals").desc())
athlete_golds.show(1)
```

```
=== 7. MAX GOLD MEDALS (Single Athlete) ===
+-----+-----+
| Athlete|Total_Gold_Medals|
+-----+-----+
```

```
+-----+-----+
|Michael Phelps|      18|
+-----+-----+
only showing top 1 row
```

8. Order athletes by total medals (descending)

```
from pyspark.sql.functions import col, sum

print("\n=== 8. ORDER ATHLETES BY TOTAL MEDALS (Descending) ===")
athlete_medals = df_full.groupBy("Athlete") \
    .agg(sum("`Total Medals`").alias("Total_Medals")) \
    .orderBy(col("Total_Medals").desc())
athlete_medals.show(5)
```

```
=== 8. ORDER ATHLETES BY TOTAL MEDALS (Descending) ===
+-----+-----+
| Athlete|Total_Medals|
+-----+-----+
| Michael Phelps|      22|
| Natalie Coughlin|      12|
| Ryan Lochte|      11|
| Leisel Jones|       9|
| Ian Thorpe|       9|
+-----+-----+
only showing top 5 rows
```

9. Number of athletes per country

```
from pyspark.sql.functions import countDistinct

print("\n=== 9. NUMBER OF ATHLETES PER COUNTRY ===")
athletes_per_country = df_full.groupBy("Country") \
    .agg(countDistinct("Athlete").alias("Number_of_Athletes"))
athletes_per_country.show(5)
```

```
=== 9. NUMBER OF ATHLETES PER COUNTRY ===
+-----+-----+
| Country|Number_of_Athletes|
+-----+-----+
| Russia|      567|
| Paraguay|      17|
| Chinese Taipei|      18|
| Sweden|     143|
| Eritrea|       1|
+-----+-----+
only showing top 5 rows
```

10. Find athletes who won at least 1 medal of each type (Gold, Silver, Bronze)

```
at_least = df_full.filter((df_full["`Gold Medals`"] >= 1) & (df_full["`Silver Medals`"] >= 1) & (df_full["`Bronze Medals`"] >= 1))
at_least.show()
at_least.count()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Athlete|Age| Country|Year|Closing Ceremony Date| Sport|Gold Medals|Silver Medals|Bronze Medals|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Natalie Coughlin| 25|United States|2008| 08-24-08| Swimming| 1| 2| 3|
| Aleksey Nemov| 24| Russia|2000| 10-01-00| Gymnastics| 2| 1| 3|
| Alicia Coutts| 24| Australia|2012| 08-12-12| Swimming| 1| 3| 1|
| Ryan Lochte| 27|United States|2012| 08-12-12| Swimming| 2| 2| 1|
| Allison Schmitt| 22|United States|2012| 08-12-12| Swimming| 3| 1| 1|
| Natalie Coughlin| 21|United States|2004| 08-29-04| Swimming| 2| 2| 1|
| Cindy Klassen| 26| Canada|2006| 02-26-06| Speed Skating| 1| 2| 2|
| Nastia Liukin| 18|United States|2008| 08-24-08| Gymnastics| 1| 3| 1|
| Marit Bjørgen| 29| Norway|2010| 02-28-10| Cross Country Skiing| 3| 1| 1|
| Sun Yang| 20| China|2012| 08-12-12| Swimming| 2| 1| 1|
| Libby Lenton-Tric...| 23| Australia|2008| 08-24-08| Swimming| 2| 1| 1|
| Inge de Bruijn| 30| Netherlands|2004| 08-29-04| Swimming| 1| 1| 2|
| Ian Thorpe| 21| Australia|2004| 08-29-04| Swimming| 2| 1| 1|
| Gary Hall Jr.| 25|United States|2000| 10-01-00| Swimming| 2| 1| 1|
| Aliya Mustafina| 17| Russia|2012| 08-12-12| Gymnastics| 1| 1| 2|
| Dmitry Sautin| 26| Russia|2000| 10-01-00| Diving| 1| 1| 2|
| Petter Northug Jr.| 24| Norway|2010| 02-28-10| Cross Country Skiing| 2| 1| 1|
| Camille Muffat| 22| France|2012| 08-12-12| Swimming| 1| 1| 1|
| Alain Bernard| 25| France|2008| 08-24-08| Swimming| 1| 1| 1|
| Kirsty Coventry| 20| Zimbabwe|2004| 08-29-04| Swimming| 1| 1| 1|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
40
```

B. Perform on players.csv and save each result in different data frame variable:

1. Read the CSV

```
player_df = spark.read.csv("players .csv",sep=",", header=True, inferSchema=True)
player_df.show()
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
3	Neymar Jr	NULL	Brazil	15	Al Hilal
4	Kylian Mbappe	25	NULL	30	PSG
5	Kevin De Bruyne	32	Belgium	NULL	Man City
6	NULL	29	France	18	PSG
7	Luka Modric	38	Croatia	10	NULL
8	NULL	NULL	NULL	NULL	NULL
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
11	Karim Benzema	36	France	NULL	Al Ittihad
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
14	Eden Hazard	33	Belgium	NULL	Real Madrid
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
20	Gavi	19	Spain	NULL	Barcelona

only showing top 20 rows

2. Show Schema

```
#Show Schema
player_df.printSchema()
```

```
root
|-- PlayerID: integer (nullable = true)
|-- Name: string (nullable = true)
|-- Age: integer (nullable = true)
|-- Country: string (nullable = true)
|-- Goals: integer (nullable = true)
|-- Club: string (nullable = true)
```

3. Count missing (null) values per column

```
#Count missing (null) values per column
from pyspark.sql.functions import col, sum

missing_values = player_df.select([sum(col(c).isNull().cast("int")).alias(c) for c in player_df.columns])
missing_values.show()
```

PlayerID	Name	Age	Country	Goals	Club
0	2	2	2	6	3

4. Drop rows with any missing values

```
#Drop rows with any missing values
player_df = player_df.dropna()
player_df.show()
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
21	Son Heung-min	32	South Korea	16	Tottenham

22	Bruno Fernandes	30	Portugal	9	Man United
23	Marcus Rashford	27	England	13	Man United
24	Jude Bellingham	21	England	17	Real Madrid
25	Phil Foden	24	England	15	Man City
26	Joao Felix	26	Portugal	6	Atletico Madrid
27	Antoine Griezmann	34	France	12	Atletico Madrid
28	Olivier Giroud	38	France	10	AC Milan

5. Drop rows only if all values are missing

```
player_df = player_df.dropna(how='all')
player_df.show()
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
21	Son Heung-min	32	South Korea	16	Tottenham
22	Bruno Fernandes	30	Portugal	9	Man United
23	Marcus Rashford	27	England	13	Man United
24	Jude Bellingham	21	England	17	Real Madrid
25	Phil Foden	24	England	15	Man City
26	Joao Felix	26	Portugal	6	Atletico Madrid
27	Antoine Griezmann	34	France	12	Atletico Madrid
28	Olivier Giroud	38	France	10	AC Milan

6. Fill missing numeric values (Goals) with 0 and Fill missing string values (Club) with "Unknown"

```
#Fill missing numeric values (Goals) with 0 and Fill missing string values (Club) with "Unknown"
from pyspark.sql.functions import when

player_df = player_df.withColumn("Goals", when(player_df["Goals"].isNull(), 0).otherwise(player_df["Goals"]))
player_df = player_df.withColumn("Club", when(player_df["Club"].isNull(), "Unknown").otherwise(player_df["Club"]))
player_df.show()
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
21	Son Heung-min	32	South Korea	16	Tottenham
22	Bruno Fernandes	30	Portugal	9	Man United
23	Marcus Rashford	27	England	13	Man United
24	Jude Bellingham	21	England	17	Real Madrid
25	Phil Foden	24	England	15	Man City
26	Joao Felix	26	Portugal	6	Atletico Madrid
27	Antoine Griezmann	34	France	12	Atletico Madrid
28	Olivier Giroud	38	France	10	AC Milan

7. Show the records of players with more than 20 goals

```
player_df.filter(player_df["Goals"] > 20).show()
```

PlayerID	Name	Age	Country	Goals	Club
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
17	Mohamed Salah	32	Egypt	21	Liverpool

```
+-----+-----+-----+-----+-----+
```

8. Find the average, maximum, and minimum goals in one go.

```
#Find the average, maximum, and minimum goals in one go.  
from pyspark.sql.functions import avg, max, min
```

```
player_df.select(avg("Goals").alias("Average_Goals"),  
                 max("Goals").alias("Max_Goals"),  
                 min("Goals").alias("Min_Goals")).show()
```

```
+-----+-----+-----+  
|      Average_Goals|Max_Goals|Min_Goals|  
+-----+-----+-----+  
|14.368421052631579|      28|        5|  
+-----+-----+-----+
```